ASSESSMENT TOOL

ASSIGNMENT

CO2 - AT1

Name	SHAIK THOUHID
Register Number	192421280
Course Code	ITA0609

Topics: Genetic Algorithm and Perceptron Model

Question 1: Genetic Algorithm

A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

Solution

The objective function is $f(x) = x^2$. In a Genetic Algorithm, the fitness value tells how good each individual is. Since the function must be maximized, a larger value of x^2 represents a better individual.

$$\text{Fitness}(x) = f(x) = x^2$$

Individual x	4-bit Binary Form	Fitness $f(x) = x^2$	Selection Rank
2	0010	4	4
4	0100	16	3
6	0110	36	2
8	1000	64	1

Best candidates selected: x = 8 and x = 6, because they have the two highest fitness values, 64 and 36 respectively.

Crossover Operation

To perform a meaningful least-significant-bit crossover, the two least significant bits are exchanged between the selected parents. Using 4-bit binary representation:

Parent 1 (8)	**10 | 00**	↔	**01 | 10**	**Parent 2 (6)**
Exchange	00	↔	10	**Exchange**
Offspring 1	**10 | 10**	=	**1010**	**Decimal 10**
Offspring 2	**01 | 00**	=	**0100**	**Decimal 4**

$$\text{Offspring 1} = 1010_2 = 10, \quad \text{Fitness} = 10^2 = 100$$

$$\text{Offspring 2} = 0100_2 = 4, \quad \text{Fitness} = 4^2 = 16$$

Note: If only the single least significant bit is exchanged, both parents end in 0, so no change occurs. Therefore, exchanging the last two bits is used here to demonstrate crossover clearly.

Role of Mutation

Mutation randomly changes one or more bits in an offspring after crossover. For example, a chromosome such as 1010 may mutate to 1011 by flipping its last bit.

- Maintains genetic diversity in the population.
- Prevents the algorithm from becoming stuck at a local optimum.
- Introduces new solutions that may not be produced by crossover alone.
- Improves exploration of the search space and can lead to a better final solution.

Mutation is normally applied with a small probability. Too much mutation makes the search random, while too little mutation may reduce diversity.

Genetic Algorithm Process

1. Initial Population	2. Fitness Evaluation	3. Selection	4. Crossover	5. Mutation

The process moves from the initial population through fitness evaluation, selection and crossover, followed by mutation to preserve diversity.

Final Result

The best individuals from the initial population are 8 and 6. After exchanging their two least significant bits, the offspring are 10 and 4. The new individual 10 has fitness 100, which is better than the highest initial fitness of 64. Mutation further helps the algorithm discover different and potentially better solutions.

Question 2: Perceptron Model

A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is –0.4. For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not.

Given Values

Term	Symbol	Value
Advertisement exposure	x_1	2
Income level	x_2	1
Weight for advertisement	w_1	0.5
Weight for income	w_2	0.3
Bias	b	–0.4

Step 1: Calculate the Net Input

The perceptron calculates the weighted sum of the inputs and adds the bias.

$$\text{Net Input} = (w_1 \times x_1) + (w_2 \times x_2) + b$$

$$= (0.5 \times 2) + (0.3 \times 1) - 0.4$$

$$= 1.0 + 0.3 - 0.4 = 0.9$$

Step 2: Apply the Step Activation Function

Condition	Output
Net input ≥ 0	1 = Purchase

Since the calculated net input is 0.9 and $0.9 \geq 0$, the step activation function produces an output of 1.

Final Answer

Net Input	**0.9**
Perceptron Output	**1 – Customer makes a purchase**

Therefore, according to the perceptron model, the customer is classified as likely to purchase the product.